OCM Meeting System — Backend Architecture & Technical Overview

> ***Target Audience:*** *Management, Technical Leads, and Stakeholders*
> ***Prepared for:*** *Office of the Council of Ministers (OCM) Document & Meeting Management System*
> ***Date:*** *November 2026*
> ***System Version:*** *Production Release*

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1. Executive Summary

The OCM Document and Meeting Management backend is an enterprise-grade, RESTful API built on **Laravel 12** and **PHP 8.4**. Designed specifically for high-security public sector workflows, it orchestrates incoming government documents, multi-stage departmental approvals, meeting agenda decisions, and accompanying dispatch letters (*ຜືກາຫຍ*).

Key Architectural Strengths

- **Service-Oriented Architecture:** Clear separation of concerns with controllers handling HTTP transport and dedicated service classes encapsulating business logic.
- **Strict Role-Based Access Control (RBAC):** Multi-tier access segmentation for Officers, Office Heads, Department Heads, and System Administrators.
- **Full Accountability (Tamper-evident Audit Trail):** Every record modification, file download, and authorization state transition is recorded with actor metadata.
- **Real-Time Workload Engine:** Dynamic, zero-cache calculation of actionable tasks sorted by waiting duration to eliminate administrative bottlenecks.
- **Automated API Documentation:** Living OpenAPI / Swagger documentation auto-generated via Dedoc Scramble.

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2. Technology Stack & Framework Choices

Layer	Component	Version / Spec	Justification
Runtime Environment	PHP	`^8.4`	Modern type-safety, performance optimizations, and memory efficiency.
Application Framework	Laravel Framework	`^12.0` / `^13.0`	Enterprise-grade MVC architecture, robust Eloquent ORM, built-in queue system, and event handling.
Authentication	Laravel Sanctum	`^4.0`	Token-based API authentication with per-device session management and remote revocation capabilities.
Authorization	Spatie Laravel Permission	`^8.3`	Industry-standard role and permission matrix mapped to organizational hierarchies.
Auditing & Logging	Spatie Laravel Activitylog	`^4.12`	Immutable event log capturing model attribute diffs, IP addresses, and user-agent context.

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Office & Reporting	PhpSpreadsheet	`^5.10`	Server-side generation and streaming of Excel (`.xlsx`) and CSV reports.
API Documentation	Dedoc Scramble	`^0.13`	Reflection-based OpenAPI schema generation without tedious PHPDoc boilerplate.
Code Quality	Laravel Pint	`^1.27`	Strict adherence to PSR-12 and modern Laravel coding conventions.

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3. Core Functional Modules

3.1 Authentication, Sessions & Security

- **Bearer Token Lifecycle:** Issues revocable personal access tokens for authenticated sessions.
- **Mandatory Password Rotation:** First-time logins are gated by the `EnsurePasswordChanged` middleware until credentials are personalized.
- **Session Supervision:** Administrators can monitor active login sessions across IP addresses and remotely terminate unauthorized or stale sessions.
- **Multi-tiered Rate Limiting:** Enforces independent throttle boundaries for login attempts, password updates, file uploads, and privileged administrative endpoints.

3.2 Incoming Documents Workflow (*ឯកសារចូល*)

- **Multi-File Attachment Processing:** Secure handling, MIME validation, and disk storage for attached government PDFs.
- **Sequential Lifecycle States:**
  1. `pendingSend` (Draft / Pending submission)
  2. `waitingReview` (Office Head examination)
  3. `waitingDecision` (Department Head meeting schedule & type decision)
  4. `needWarrant` (Officer dispatch preparation)
  5. `completed` (Concluded with finalized accompanying letter)
- **Attachment Tracking:** Granular logging of file streaming and download confirmations.

3.3 Accompanying Letters (*ដីកាអម*)

- **Sequential Number Generator (`next-number`):** Concurrency-safe algorithm ensuring unique, ordered dispatch numbers by calendar year.
- **Attendee & Ministry Association:** Relates external ministry participants, agendas, and venue decisions directly to the dispatch record.
- **Print State Tracking:** Records timestamped print acknowledgments to prevent duplicate physical issuances.

3.4 Member Groups & Institutional Directory

- **Hierarchical Directory:** Maintains structured ministries, secretariats, and departmental units.
- **Active / Deleted Roster:** Manages personnel contacts with soft deletion and audit-logged membership adjustments.

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3.5 Real-Time Summary Dashboard (*ផ្ទាំងសង្ខេប*)

- **Oldest-Waiting-First Algorithm:** Automatically surfaces the top 6 most urgent documents waiting for individual action based on `tracked_at` timestamps.
- **Acting Officer Delegation:** Seamlessly aggregates workloads if a user has an active acting assignment (*ស្តីទី*).
- **Personal Year-to-Date Insights:** Summarizes individual contributions (approved documents, scheduled sessions, authored letters) via real-time activity log aggregation.

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4. Security Architecture & Middleware Pipeline

Every HTTP request traverses a multi-layered middleware pipeline prior to execution:

1. `SecurityHeaders`: Applies strict HTTP headers (X-Frame-Options, X-Content-Type-Options, Referrer-Policy, CSP directives).
2. `DetectSuspiciousInput`: Inspects input bodies for payload anomalies, SQL injection vectors, and malformed script patterns.
3. `throttle:*`: Prevents brute-force attacks on sensitive auth routes (`throttle:login`, `throttle:password-reset`) and resource exhaustion on file endpoints (`throttle:uploads`).
4. `auth:sanctum`: Validates the cryptographic signature of the Bearer token against active tokens in the database.
5. `EnsurePasswordChanged`: Rejects access to application routes if the account is flagged with a temporary initial password.
6. `EnsureMenuSection`: Validates whether the caller's role possesses explicit navigational permission (`documents`, `letters`, `names`).
7. `EnsureUserIsAdmin`: Restricts the `/api/v1/admin/*` surface to verified administrative actors.

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5. Database Schema & Index Optimization

The database is structured to balance strict referential integrity with sub-millisecond query performance:

- **Composite Performance Indexes:** High-frequency query columns are indexed in pairs (e.g., `(role, is_active)` on `users`, `(status, deleted_at, tracked_at)` on `documents`, `(log_name, causer_id, created_at)` on `activity_log`).
- **Normalized Relationships:** Foreign key constraints with cascading logic enforce parent-child consistency across documents, attachment files, and accompanying letters.
- **Audit Immutability:** Audit records remain decoupled from entity deletions to guarantee legal and regulatory compliance.

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6. Recommended Presentation Flow

When presenting this architecture to stakeholders, structure your walkthrough as follows:

Order	Section	Recommended Talking Points

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1	**System Objectives**	Built to modernize OCM document handling, remove paper bottlenecks, and enforce accountability.
2	**Tech Stack & Standards**	Modern PHP 8.4 + Laravel 12 foundation with PSR-12 code styling and auto-generated API docs.
3	**Access & Security**	Token-based auth, forced password change on first login, role menus, and rate limiting.
4	**Document Lifecycle**	Walk through the journey of a document from receipt to meeting decision to accompanying letter.
5	**Executive Dashboard**	Show how officers receive prioritized, real-time queues without manual filtering.
6	**Audit Trail & Compliance**	Demonstrate the complete log of changes and Excel export capabilities.